

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286226

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

WU; Kai et al.

BATTERY CELL, BATTERY, AND ELECTRICAL DEVICE

Abstract

The battery cell comprises an outer housing, an electrode assembly, and an insulating member. The end of the outer housing in the first direction (Z) is provided with a pressure relief mechanism. The insulating member is provided in the outer housing and is located at the end of the electrode assembly near the pressure relief mechanism. In a direction intersecting the first direction (Z), a pressure relief gap is provided between a side portion of the insulating member and the outer housing. A pressure relief recess in communication with the pressure relief gap is disposed on the side portion of the insulating member. A pressure relief channel that is in communication with the pressure relief recess and disposed opposite to the pressure relief mechanism is also provided in the outer housing.

Inventors: WU; Kai (Ningde, CN), LI; Yao (Ningde, CN), LIU; Sike (Ningde, CN), CHEN; Shilong (Ningde, CN), GU; Mingguang (Ningde, CN)

Applicant: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

Family ID: 87751392

Assignee: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

Appl. No.: 19/216105

Filed: May 22, 2025

Foreign Application Priority Data

CN

202321141617.6

May. 12, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2023/109917 20230728 PENDING child US 19216105

Publication Classification

Int. Cl.: H01M50/477 (20210101); B60L50/64 (20190101); H01M50/103 (20210101); H01M50/342 (20210101); H01M50/474 (20210101)

U.S. Cl.:

CPC H01M50/477 (20210101); H01M50/103 (20210101); H01M50/3425 (20210101); H01M50/474 (20210101); B60L50/64 (20190201); H01M2220/20 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International application PCT/CN2023/109917 filed on Jul. 28, 2023 that claims priority to Chinese Patent Application No. 202321141617.6 filed on May 12, 2023. The content of these applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of batteries, and in particular, to a battery cell, a battery, and an electric device.

BACKGROUND

[0003] In the related art, a battery cell includes an electrode assembly, a housing body, an end cover, and an insulating member. The electrode assembly and the insulating member are both accommodated in an interior environment defined by the housing and end cover. The insulating member is disposed between the electrode assembly and the end cover to insulate the electrode assembly from the end cover.

[0004] In some cases, when the electrode assembly experiences thermal runaway, high-temperature and high-pressure gas will appear at the side portion of the electrode assembly. The high-temperature and high-pressure gas at the side portion of the electrode assembly is difficult to be released to a pressure relief mechanism at the end cover timely, such that the pressure in the battery cell is difficult to be released timely, which is likely to result in damage or even explosion of the electrode assembly.

SUMMARY

[0005] In view of the above problem, embodiments of the present application provide a battery cell, a battery, and an electric device, which can ameliorate the technical problem of difficulties in timely releasing the gas at the side portion of the electrode assembly.

[0006] Technical solutions employed in the embodiments of the present application are as follows.

[0007] In a first aspect, an embodiment of the present application provides a battery cell. The battery cell includes: [0008] a housing provided with a pressure relief mechanism at an end in a first direction; [0009] an electrode assembly disposed in the housing; and [0010] an insulating member disposed in the housing and located at an end of the electrode assembly proximal to the pressure relief mechanism, where in a direction intersecting with the first direction, a pressure relief gap is provided between a side portion of the insulating member and the housing, the insulating member is provided, at the side portion, with a pressure relief slot in communication with the pressure relief gap, and the housing is provided therein with a pressure relief channel in communication with the pressure relief slot and disposed opposite to the pressure relief mechanism. [0011] In the battery cell provided in the embodiments of the present application, in the direction intersecting with the first direction, the pressure relief gap is provided between the side portion of the insulating member and the housing, the insulating member is provided with the pressure relief

slot at the side portion, the housing is provided therein with the pressure relief channel opposite to the pressure relief mechanism, and the pressure relief slot is in communication with the pressure relief gap and the pressure relief channel. In this way, when the electrode assembly experiences thermal runaway, the high-temperature and high-pressure gas at the side portion of the electrode assembly can sequentially pass through the pressure relief gap and the pressure relief slot to flow to the pressure relief channel and thus to the pressure relief mechanism, such that the gas can be released through the pressure relief mechanism. In such a configuration, the high-temperature and high-pressure gas at the side portion of the electrode assembly can be released to the pressure relief mechanism timely to achieve timely pressure relief, which can ameliorate the problem of damage or even explosion of the electrode assembly caused by the difficulties in timely pressure relief of the electrode assembly.

[0012] In some embodiments, the pressure relief channel is provided penetrating through the insulating member in the first direction.

[0013] In such a configuration, the release of the high-temperature and high-pressure gas can be effectively achieved to achieve timely pressure relief of the battery cell.

[0014] In some embodiments, the insulating member includes a body part and a first protrusion connected to the body part, where the first protrusion extends beyond the body part in the first direction; the pressure relief gap includes a first gap provided between the first protrusion and the housing, the pressure relief slot includes a first slot provided in the first protrusion and in communication with the first gap, the pressure relief channel includes a first channel provided in the first protrusion, and the first channel is in communication with the first slot and is opposite to the pressure relief mechanism.

[0015] In such a configuration, the release path of the high-temperature and high-pressure gas can be shortened to quickly realize the pressure relief of the high-temperature and high-pressure gas, thereby ameliorating the problem of difficulties in timely releasing the high-temperature and high-pressure gas caused by the first protrusion blocking the high-temperature and high-pressure gas.

[0016] In some embodiments, the first protrusion and the body part are sequentially disposed in a second direction; in a third direction, the first gap is provided between a side portion of the first protrusion and the housing, and the first protrusion is provided with the first slot at the side portion; the first direction, the second direction, and the third direction intersect with each other, and the first direction, the second direction, and the third direction are not in a same plane.

[0017] In such a configuration, the release of the high-temperature and high-pressure gas at the side portion of the electrode assembly in the third direction can be effectively achieved to achieve timely pressure relief of the battery cell.

[0018] In some embodiments, the first protrusion includes: [0019] two first walls spaced apart from each other on the body part in the second direction to form the first channel, the first slot being formed in side portions of the two first walls in the third direction.

[0020] By forming the first channel by the two first walls' being spaced apart from each other in the second direction, the first channel may be provided penetrating through the insulating member in the first direction, and the first channel has a larger dimension in a cross section perpendicular to the first direction, which facilitates the high-temperature and high-pressure gas's passing through the first channel to flow to the pressure relief mechanism and thereby to achieve timely pressure relief of the battery cell. Also, the ends of the two first walls in the third direction may be spaced apart to form the first slot, such that the first slot has a large dimension, which also facilitates the passing of the high-temperature and high-pressure gas at the side portion of the electrode assembly in the third direction through the first gap and the first slot to enter the first channel, thus facilitating the timely pressure relief of the electrode assembly.

[0021] In some embodiments, the first protrusion further includes: [0022] a second wall disposed between the two first walls and at the side portions of the first walls in the third direction, the first walls and the second wall defining the first channel, and the first slot being formed in the second

wall.

[0023] With the technical solution described above, the second wall is provided between the two first walls, and the first slot is formed in the second wall. As such, the insulating film may extend to the second wall to be fixed to the second wall, such that the insulating film remains in the standing state, thereby maintaining the insulation effect between the electrode assembly and the housing.

[0024] In some embodiments, the first protrusion further includes a third wall, the third wall being disposed between the two first walls and opposite to the pressure relief mechanism; the first channel includes: [0025] a first space in communication with the first slot and opposite to the pressure relief mechanism, the first walls and a side of the third wall distal to the electrode assembly in the first direction defining the first space; and [0026] second slots penetrating through the third wall in the first direction and in communication with the first space.

[0027] In such a configuration, the high-temperature and high-pressure gas can be buffered in the first space and then flow to the pressure relief mechanism, allowing the high-temperature and high-pressure gas in the battery cell to rush out through the pressure relief mechanism to realize pressure relief.

[0028] In some embodiments, the third wall is disposed at ends of the first walls facing the electrode assembly in the first direction and is configured to abut against the electrode assembly in the first direction.

[0029] With the technical solution described above, the third wall may abut against the electrode assembly in the first direction. As such, the area of the first protrusion for contact with the electrode assembly can be increased, such that the first protrusion can abut against the electrode assembly very stably, thus fixing the electrode assembly in the housing and allowing for reduced damages to the electrode assembly caused by the first protrusion.

[0030] In some embodiments, the body part includes a plurality of sub-parts sequentially disposed in the second direction, and the first protrusion is disposed between two adjacent sub-parts.

[0031] In such a configuration, the first protrusion is disposed in the middle of the body part in the second direction, and the first protrusion is thus directly opposite to the middle of the electrode assembly in the second direction. As such, the high-temperature and high-pressure gas generated by the electrode assembly can quickly flow to the first channel to be released through the pressure relief mechanism. This helps to achieve the effect of timely pressure relief of the battery cell.

[0032] In some embodiments, the first slot and the first protrusion satisfy the following relationship:

$S1 \geq S2 \times 1/6$, [0033] where on one of the side surfaces of the first protrusion in the third direction, $S1$ is an area of the first slot, and $S2$ is an area of the first protrusion.

[0034] By setting $S1 \geq S2 \times 1/6$, the area of the first slot is not too small, such that the first slot can allow the high-temperature and high-pressure gas to flow through timely, which facilitates the high-temperature and high-pressure gas' entering the first channel through the first slot for pressure relief.

[0035] In some embodiments, $S1 \geq S2 \times 1/2$.

[0036] By setting $S1 \geq S2 \times 1/2$, the first slot can have a large area proportion on the first protrusion, such that the effect of rapid release of the high-temperature and high-pressure gas at the side portion of the electrode assembly to the pressure relief mechanism can be achieved.

[0037] In some embodiments, the first protrusion is disposed on a side portion of the body part in the second direction; in the second direction, the first gap is provided between a side portion of the first protrusion distal to the body part and the housing, and the first protrusion is provided with the first slot at the side portion; the first direction intersects with the second direction.

[0038] In such a configuration, the high-temperature and high-pressure gas on the side portion of the electrode assembly in the second direction can sequentially pass through the first gap and the first slot to enter the first channel and thereby to flow to the pressure relief mechanism through the

first channel. This enables the effect of timely release of the high-temperature and high-pressure gas at the side portion of the electrode assembly in the second direction to achieve timely pressure relief of the electrode assembly.

[0039] In some embodiments, the insulating member includes the body part and a second protrusion; in the first direction, the second protrusion is disposed on a side of the body part facing the electrode assembly, and the body part is spaced apart from the electrode assembly to form a second space in communication with the pressure relief channel; the pressure relief gap includes a second gap, and the pressure relief slot includes a third slot in communication with the second gap; in a direction intersecting with the first direction, the second protrusion is spaced apart from the housing to form the second gap, and the second protrusion is provided with the third slot in communication with the second space.

[0040] With the technical solution described above, in the case where there is high-temperature and high-pressure gas at the side portion of the electrode assembly in the direction intersecting with the first direction, the high-temperature and high-pressure gas can sequentially pass through the first gap and the second gap to flow to the pressure relief channel, or can pass through the second gap and the third slot to flow to the second space and then to the pressure relief channel, thus being released to the pressure relief mechanism. This helps to achieve the effect of timely pressure relief of the electrode assembly.

[0041] In some embodiments, the insulating member includes the body part and the first protrusion, and the first protrusion and the body part are sequentially disposed in the second direction; in the first direction, the first protrusion extends beyond the body part toward the electrode assembly, and the body part is spaced apart from the electrode assembly to form the second space in communication with the pressure relief channel; the pressure relief channel includes the first channel provided in the first protrusion and opposite to the pressure relief mechanism, and the first protrusion is provided, at a side portion in the second direction, with fourth slots in communication with the second space and the first channel; the first direction intersects with the second direction.

[0042] In such a configuration, the high-temperature and high-pressure gas at the side portion of the electrode assembly in the direction intersecting with the first direction can flow to the second space through a plurality of pathways, and thus flow to the first channel through the fourth slots to be released through the pressure relief mechanism, such that the battery cell has a plurality of channels that can release the high-temperature and high-pressure gas, which contributes to the timely pressure relief of the battery cell.

[0043] In some embodiments, the insulating member includes the body part, and the body part is spaced apart from the electrode assembly in the first direction to form the second space; the pressure relief channel includes a second channel, and the second channel includes: [0044] a fifth groove provided on a side of the body part distal to the electrode assembly in the first direction and opposite to the pressure relief mechanism; and [0045] sixth slots penetrating through a bottom of the fifth groove in the first direction and in communication with the second space.

[0046] In such a configuration, the high-temperature and high-pressure gas in the second space may not only flow to the first channel through the fourth slots, but also flow to the sixth slots through the fifth groove, and is then released to the pressure relief mechanism. In this way, the high-temperature and high-pressure gas in the second space can be released to the pressure relief mechanism through a plurality of pathways, thus facilitating the timely pressure relief of the battery cell.

[0047] In a second aspect, an embodiment of the present application provides a battery. The battery includes a battery cell.

[0048] According to the battery provided in the embodiment of the present application, the use of the battery cell referred to above can ameliorate the problem of difficulties in timely releasing the high-temperature and high-pressure gas from the electrode assembly caused by the insulating

member blocking the high-temperature and high-pressure gas, so as to ameliorate the problem of damage or even explosion of the electrode assembly caused by the difficulties in timely pressure relief of the electrode assembly, thereby imparting the battery the advantage of timely pressure relief.

[0049] In a third aspect, an embodiment of the present application provides an electric device. The electric device includes a battery cell or a battery.

[0050] According to the electric device provided in the embodiment of the present application, the use of the battery cell or battery referred to above can ameliorate the problem of difficulties in timely releasing the high-temperature and high-pressure gas from the electrode assembly caused by the insulating member blocking the high-temperature and high-pressure gas, so as to ameliorate the problem of damage or even explosion of the electrode assembly caused by the difficulties in timely pressure relief of the electrode assembly, thereby imparting the battery the advantage of timely pressure relief.

[0051] The above description is only an overview of the technical solutions of the present application. To more clearly understand the technical means of the present application to enable implementation in accordance with the content of the specification and to make the above and other purposes, features, and advantages of the present application more obvious and easy to understand, the detailed description of the present application is provided below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0052] To more clearly illustrate the technical solutions in the embodiments of the present application, the drawings required for illustrating the embodiments or exemplary technology are briefly described below. Apparently, the drawings in the following description illustrate merely some embodiments of the present application, and those of ordinary skill in the art may still obtain other drawings based on these drawings without creative efforts.

[0053] FIG. 1 is a schematic diagram of a vehicle according to some embodiments of the present application;

[0054] FIG. 2 is a schematic exploded view of a battery according to some embodiments of the present application;

[0055] FIG. 3 is a schematic diagram of a battery cell according to some embodiments of the present application;

[0056] FIG. 4 is an exploded view of FIG. 3;

[0057] FIG. 5 is a schematic perspective view of an insulating member of the battery cell provided in FIG. 3;

[0058] FIG. 6 is a partial enlarged view of FIG. 5;

[0059] FIG. 7 is a cross-sectional view of FIG. 3 along line A-A;

[0060] FIG. 8 is an enlarged view of part B in FIG. 7;

[0061] FIG. 9 is a schematic perspective view of an insulating member of a battery cell according to some other embodiments of the present application;

[0062] FIG. 10 is a cross-sectional view of the battery cell where the insulating member provided in FIG. 9 is located;

[0063] FIG. 11 is an enlarged view of part C in FIG. 10;

[0064] FIG. 12 is an enlarged view of part D in FIG. 10; and

[0065] FIG. 13 is a partial enlarged view of FIG. 9.

[0066] Reference numerals in the drawings have the following meanings:

[0067] **1000**—vehicle; **100**—battery; **200**—controller; **300**—motor; **10**—battery cell; **20**—case; **21**—first part; **22**—second part; **101**—pressure relief gap; **1011**—first gap; **1012**—second gap; **102**—

pressure relief slot; **1021**—first slot; **1022**—third slot; **103**—pressure relief channel; **1031**—first channel; **10311**—first space; **10312**—second slot; **1032**—second channel; **10321**—fifth groove; **10322**—sixth slot; **104**—second space; **105**—fourth slot; **11**—electrode assembly; **12**—housing; **121**—housing body; **122**—end cover; **123**—pressure relief mechanism; **13**—insulating member; **131**—body part; **1311**—sub-part; **132**—first protrusion; **1321**—first wall; **1322**—second wall; **1323**—third wall; **133**—second protrusion; **14**—insulating film; Z-first direction; Y-second direction; X-third direction.

DETAILED DESCRIPTION

[0068] The embodiments of the present application are described in detail hereinafter, with examples of the embodiments illustrated in the drawings. Throughout the drawings, the same or similar reference numbers indicate the same or similar elements or elements having the same or similar functions. The embodiments described hereinafter with reference to the drawings are exemplary and are intended to explain the present disclosure. They should not be construed as limiting the present disclosure.

[0069] In the description of the present application, it should be understood that the terms “length”, “width”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, and the like indicate orientations or positional relationships based on those shown in the drawings. They are merely for the convenience of describing the present application and simplifying the description, rather than indicating or implying that the apparatus or element referred to must have a specific orientation or be constructed and operate in the specific orientation, and thus should not be construed as a limitation to the present application.

[0070] In addition, the terms “first” and “second” are used for descriptive purposes only and should not be construed as indicating or implying relative importance or implicitly indicating the number of technical features described. Therefore, a feature defined with “first” or “second” may explicitly or implicitly include one or more such a feature.

[0071] In the description of the present application, “a plurality of” means no less than two, and “no less than two” includes two, unless otherwise specifically defined. Correspondingly, “a plurality of groups” means no less than two groups.

[0072] In the description of the present application, unless otherwise clearly specified and defined, the terms “mount”, “interconnect”, “connect”, “fix”, and the like should be interpreted in their broad senses. For example, “connect” may be “fixedly connect”, “detachably connect”, or “integrally connect”; “mechanically connect” or “electrically connect”; or “directly interconnect”, “indirectly interconnect through an intermediate”, “communication between interiors of two elements”, or “interaction between two elements”. For those of ordinary skill in the art, the specific meanings of the aforementioned terms in the present application can be understood according to specific conditions.

[0073] In the description of the present application, the term “and/or” is merely a way to describe the associative relationship between associated objects, indicating that there are three possible relationships. For example, “A and/or B” may denote: the presence of A alone, the presence of both A and B, and the presence of B alone. In addition, the character “/” herein generally indicates an “or” relationship between the associated objects before and after the “/”.

[0074] Although the present application has been described with reference to preferred embodiments, various modifications can be made and components can be replaced with equivalents without departing from the scope of the present application. In particular, the technical features mentioned in the embodiments may be combined in any manner as long as there are no structural conflicts. The present application is not limited to the specific embodiments disclosed herein, but encompasses all technical solutions falling within the scope of the claims.

[0075] In the related art, a battery cell includes an electrode assembly, a housing body, an end cover, and an insulating member. The electrode assembly and the insulating member are both accommodated in an interior environment defined by the housing and end cover. The insulating

member is disposed between the electrode assembly and the end cover to insulate the electrode assembly from the end cover.

[0076] In some cases, when the electrode assembly experiences thermal runaway, high-temperature and high-pressure gas will appear at the side portion of the electrode assembly. The high-temperature and high-pressure gas at the side portion of the electrode assembly is difficult to be released to a pressure relief mechanism at the end cover timely, such that the pressure in the battery cell is difficult to be released timely, which is likely to result in damage or even explosion of the electrode assembly.

[0077] Illustratively, in some cases, the insulating member has a protrusion formed through the protruding of a side thereof facing the electrode assembly. The protrusion is configured to abut against and hold an end of the electrode assembly to fix the electrode assembly. However, when the electrode assembly experiences thermal runaway, high-temperature and high-pressure gas at the side portion of the electrode assembly is blocked by the protrusion, making it difficult to release the high-temperature and high-pressure gas from the insulating member to a pressure relief mechanism timely and thereby causing hysteresis in the pressure relief of the electrode assembly.

[0078] In view of the above, embodiments of the present application provide a battery cell, a battery, and an electric device. In a direction intersecting with a first direction, a pressure relief gap is provided between the side portion of the insulating member and the housing, the insulating member is provided with a pressure relief slot at the side portion, the housing is provided therein with a pressure relief channel opposite to the pressure relief mechanism, and the pressure relief slot is in communication with the pressure relief gap and the pressure relief channel. In this way, when the electrode assembly experiences thermal runaway, the high-temperature and high-pressure gas at the side portion of the electrode assembly can sequentially pass through the pressure relief gap and the pressure relief slot to flow to the pressure relief channel and thus to the pressure relief mechanism, such that the gas can be released through the pressure relief mechanism. In such a configuration, the high-temperature and high-pressure gas at the side portion of the electrode assembly can be released to the pressure relief mechanism timely to achieve timely pressure relief, which can ameliorate the problem of damage or even explosion of the electrode assembly caused by the difficulties in timely pressure relief of the electrode assembly.

[0079] In some embodiments, the battery cell referred to in the embodiments of the present application can be used in electric devices that use battery cells or batteries as their power source, and the battery referred to in the embodiments of the present application can be used in electric devices that use batteries as their power source.

[0080] The electric device may be, but is not limited to, a mobile phone, a tablet, a laptop computer, an electric toy, an electric tool, an electric bike, an electric car, a ship, a spacecraft, and the like. The electric toy may include a stationary or mobile electric toy, such as a game console, an electric car toy, an electric ship toy, and an electric airplane toy, and the spacecraft may include an airplane, a rocket, a space shuttle, a spaceship, and the like. The vehicle may be a fuel vehicle, a gas vehicle, or a new energy vehicle. The new energy vehicle may be a pure electric vehicle, a hybrid vehicle, an extended-range vehicle, or the like.

[0081] In some embodiments, the battery cell and the battery referred to in the embodiments of the present application may also be used in an energy storage device. The energy storage device may be an energy storage container, an energy storage cabinet, or the like.

[0082] The battery referred to in the embodiments of the present application may be a single physical module including one or more battery cells to provide a higher voltage and capacity. When there are a plurality of battery cells, the plurality of battery cells are connected in series, in parallel, or in series-parallel by a busbar component. The series-parallel connection means that both series connection and parallel connection are present for the connection among the plurality of battery cells.

[0083] In some embodiments, the battery may be a battery module. When there are a plurality of

battery cells, the plurality of battery cells are arranged and fixed to form a battery module. As an example, the plurality of battery cells may be fixed by a banding strap or the like to form a battery module. As an example, the plurality of battery cells may also be fixed by an end plate, a side plate, or the like to form a battery module.

[0084] In some embodiments, the battery may be a battery pack, which may include a case and battery cells. As an example, the battery cells may be accommodated directly in the case. As an example, the battery cells may also be formed into a battery module before being accommodated in the case.

[0085] For ease of description, in the embodiments of the present application, the description is provided by taking a vehicle as an example of the electric device.

[0086] Referring to FIG. 1, FIG. 1 is a schematic diagram of a vehicle **1000** according to some embodiments of the present application. A battery **100** described above is provided inside the vehicle **1000**, and the battery **100** may be disposed at the bottom, the head, or the tail of the vehicle **1000**. The battery **100** may be configured to power the vehicle **1000**. For example, the battery **100** may serve as an operation power source of the vehicle **1000**. The vehicle **1000** may also include a controller **200** and a motor **300**. The controller **200** is used for controlling the battery **100** to power the motor **300**, e.g., for operation power needed by the vehicle **1000** for start-up, navigation, and driving.

[0087] In some embodiments, the battery **100** may not only serve as an operation power source for the vehicle **1000**, but also as a driving power source for the vehicle **1000** to, instead of or in part instead of fuel or natural gas, provide driving power for the vehicle **1000**.

[0088] Referring to FIG. 2, FIG. 2 is an exploded view of a battery **100** according to some embodiments of the present application. The battery **100** includes a case **20** and a plurality of battery cells **10**. The case **20** is a structure having an accommodating space therein, and the case **20** may be of various structures. In some embodiments, the case **20** may include a first part **21** and a second part **22**. The first part **21** and the second part **22** are mutually lidded with each other, and jointly define the above accommodating space. The first part **21** may be a hollow structure with one end open, and the second part **22** is a plate-like structure. The second part **22** lids the open side of the first part **21**, such that the first part **21** and the second part **22** jointly define the above accommodating space. Alternatively, the first part **21** and the second part **22** may also each be a hollow structure with one end open, as shown in FIG. 2, and the open side of the first part **21** lids the open side of the second part **22**, such that the first part **21** and the second part **22** jointly define the above accommodating space. The case **20** formed by the first part **21** and the second part **22** may be in various shapes such as cylindrical and rectangular parallelepiped.

[0089] In some embodiments, the plurality of battery cells **10** may be connected in series, in parallel, or in series-parallel as an integrated unit, and then the integrated unit formed by the plurality of battery cells **10** is directly accommodated in the above accommodating space of the case **20**, as shown in FIG. 2. The plurality of battery cells **10** may also be first connected in series, in parallel, or in series-parallel and then arranged and fixed to form a plurality of battery modules, the plurality of battery modules are then connected in series, in parallel, or in series-parallel to form an integrated unit, and then the integrated unit is accommodated in the above accommodating space of the case **20**.

[0090] In some embodiments, referring to FIGS. 1 and 2, when the battery **100** is applied to the vehicle **1000**, the case **20** of the battery **100** may be part of the chassis structure of the vehicle **1000**. For example, a part of the case **20** may be at least a part of the chassis of the vehicle **1000**, or a part of the case **20** may be at least a part of a transverse beam and a longitudinal beam of the vehicle **1000**.

[0091] The battery cell **10** refers to the smallest unit that stores and outputs electric energy. The battery cell **10** may be a secondary battery or a primary battery. The battery cell **10** may be, but is not limited to, a metal battery, a lithium sulfur battery, a sodium-ion battery, or a magnesium-ion

battery. The battery cell **10** may be cylindrical, flat, rectangular parallelepiped, or in other shapes. [0092] Referring to FIGS. **3** and **4** together, FIG. **3** is a schematic diagram of a battery cell **10** according to some embodiments of the present application, and FIG. **4** is an exploded view of FIG. **3**. The battery cell **10** includes an electrode assembly **11** and a housing **12**.

[0093] The electrode assembly **11** is a component where the electrochemical reaction occurs in the battery cell **10**. The electrode assembly **11** is mainly formed by winding or stacking a positive electrode plate and a negative electrode plate, and a separator is provided between the positive electrode plate and the negative electrode plate. The portions of the positive electrode plate and the negative electrode plate that contain active substances form the main body of the electrode assembly **11**, and the portions of the positive electrode plate and the negative electrode plate that do not contain active substances each form a tab. The tab of the positive electrode plate is a positive tab and the tab of the negative electrode plate is a negative tab. The positive electrode tab and the negative electrode tab may be located together at one end of the main body or respectively at two opposite ends of the main body.

[0094] In the battery cell **10**, the number of electrode assemblies **11** may be one or more.

[0095] In some cases, the electrode assembly **11** may also be referred to as a bare cell, a wound body, a stack, etc.

[0096] In some embodiments, the battery cell **10** may further include an electrolyte that conducts ions between the positive electrode plate and the negative electrode plate. In the embodiments of the present application, the electrolyte may be liquid, gel, or solid.

[0097] The housing **12** includes a housing body **121** and an end cover **122**. The housing body **121** and the end cover **122** are components for jointly defining the internal environment of the battery cell **10**. The internal environment defined by the housing body **121** and the end cover **122** is used for accommodating the electrode assembly **11** and the electrolyte. The housing body **121** and the end cover **122** may be separate components. Specifically, the housing body **121** has an opening, the end cover **122** lids the opening of the housing body **121** to define the internal environment of the battery cell **10** together with the housing body **121** and to isolate the internal environment of the battery cell **10** from the external environment. The housing body **121** and the end cover **122** may also be of an integrated structure. Specifically, a common connection surface may be formed between the end cover **122** and the housing body **121** before the electrode assembly **11** is placed into the housing, and after the electrode assembly **11** is placed into the housing, the end cover **122** lids the housing body **121** when the electrode assembly **11** needs to be encapsulated.

[0098] The number of end covers **122** may be one. The number of end covers **122** may also be two, and the two end covers **122** are respectively disposed at two opposite ends of the housing body **121**.

[0099] The housing body **121** may be cylindrical, square, etc., which may be determined depending on the specific shape and size of the electrode assembly **11**. Also, the housing body **121** and the end cover **122** may be made of a variety of materials, such as copper, iron, aluminum, stainless steel, aluminum alloy, and plastic.

[0100] Referring to FIGS. **3-8** together and in combination with the other figures, FIG. **5** shows a schematic perspective view of an insulating member **13** according to some embodiments of the present application, FIG. **6** shows a partial enlarged view of FIG. **5**, FIG. **7** shows a cross-sectional view of FIG. **3** along line A-A, and FIG. **8** is a partial enlarged view of FIG. **7**. The battery cell **10** provided in the embodiments of the present application includes a housing **12**, an electrode assembly **11**, and an insulating member **13**. The housing **12** provided with a pressure relief mechanism **123** at an end in the first direction Z. The electrode assembly **11** is disposed in the housing **12**. The insulating member **13** is disposed in the housing **12** and is located at an end of the electrode assembly **11** proximal to the pressure relief mechanism **123**. In a direction interacting with the first direction Z, a pressure relief gap **101** is provided between a side portion of the insulating member **13** and the housing **12**, the insulating member **13** is provided with a pressure

relief slot **102** at the side portion, and the pressure relief slot **102** is in communication with the pressure relief gap **101**. The housing **12** is further provided therein with a pressure relief channel **103**, and the pressure relief channel **103** is in communication with the pressure relief slot **102** and disposed opposite to the pressure relief mechanism **123**.

[0101] The pressure relief mechanism **123** is a component capable of relieving the internal pressure of the battery cell **10** when the internal pressure or temperature of the battery cell **10** reaches a threshold value. For example, when the battery cell **10** operates normally, the gas pressure inside the battery cell **10** is smaller than the activation pressure value for the pressure relief mechanism **123**, the pressure relief mechanism **123** is in an off state, and the gas inside the battery cell **10** is not in communication with the gas outside. When the battery cell **10** experiences thermal runaway under the actions of internal and external factors such as overcharge, overheat, and mechanical impact, a large amount of high-temperature and high-pressure gas is generated inside the battery cell **10**, the pressure inside the battery cell **10** is greater than the activation pressure value for the pressure relief mechanism **123**, the pressure relief mechanism **123** is changed from the off state to an on state, and the high-temperature and high-pressure gas inside the battery cell **10** can be discharged outside the battery cell **10** through the pressure relief mechanism **123**.

[0102] The pressure relief mechanism **123** may be a weak structure disposed on the housing **12**, or the pressure relief mechanism **123** may also be a pressure valve or the like. When the pressure relief mechanism **123** is a weak structure, the pressure relief mechanism **123** has a lower structural strength than other parts of the housing **12**. In this way, in the event of the thermal runaway of the battery cell **10**, the high-temperature and high-pressure gas generated by the battery cell **10** can break through the pressure relief mechanism **123** to be released outside the battery cell **10**. As an example, the pressure relief mechanism **123** may be formed integrally with the housing **12**. For example, the pressure relief mechanism **123** is a score provided in the housing **12**. As an example, the pressure relief mechanism **123** may also be provided separately and connected to the housing **12**.

[0103] The pressure relief mechanism **123** being disposed in the housing **12** may be that the pressure relief mechanism **123** is disposed in the housing body **121** or the pressure relief mechanism **123** is disposed on the end cover **122**. As an example, as shown in FIGS. **3** and **4**, the housing **12** includes a housing body **121** and an end cover **122**. The end cover **122** is disposed at an end of the housing body **121** in the first direction Z, and the pressure relief mechanism **123** is disposed on the end cover **122**.

[0104] The insulating member **13** refers to a component with insulating properties. Specifically, the insulating member **13** is located in the housing **12** and can insulate the electrode assembly **11** from the end of the housing **12** where the pressure relief mechanism **123** is provided. The insulating member **13** may be made of a material such as, but not limited to, plastic.

[0105] The insulating member **13** being located at the end of the electrode assembly **11** proximal to the pressure relief mechanism **123** means that the insulating member **13** is disposed at the end of the electrode assembly **11** facing toward the pressure relief mechanism **123** in the first direction Z. It may be appreciated that the insulating member **13** is disposed between the electrode assembly **11** and the pressure relief mechanism **123** in the first direction Z. In some possible designs, as shown in FIGS. **3** and **4**, the housing **12** is provided with the pressure relief mechanism **123** at one end in the first direction Z. Correspondingly, the pressure relief mechanism **123** is disposed at one end of the electrode assembly **11** in the first direction Z. Based on this, in the first direction Z, the insulating member **13** is disposed between the electrode assembly **11** and the pressure relief mechanism **123**. In some possible designs, the housing **12** is provided with a pressure relief mechanisms **123** at each of two opposite ends in the first direction Z. Correspondingly, the electrode assembly **11** has the pressure relief mechanism **123** at each of two opposite ends in the first direction Z. Based on this, the insulating member **13** may be disposed at either end of the electrode assembly **11** in the first direction Z. Alternatively, the electrode assembly **11** is provided

with an insulating member **13** at each of two opposite ends in the first direction Z.

[0106] It should be noted that the housing **12** is provided with the pressure relief channel **103**, and the pressure relief channel **103** is in communication with the pressure relief slot **102** and disposed opposite to the pressure relief mechanism **123** in the first direction Z. Based on this, the gas in the pressure relief channel **103** can flow to the pressure relief mechanism **123**. In some possible designs, the insulating member **13** and the pressure relief mechanism **123** are spaced apart from each other to form the pressure relief channel **103** described above. In some possible designs, the pressure relief channel **103** described above is formed in the insulating member **13**. In some possible designs, the pressure relief channel **103** described above is formed in the insulating member **13** and between the insulating member **13** and the pressure relief mechanism **123**.

[0107] It should also be noted that the pressure relief gap **101** is provided between a side portion of the insulating member **13** in the direction intersecting with the first direction Z and an inner side wall of the housing **12**. The insulating member **13** is provided with the pressure relief slot **102** at the side portion in the direction intersecting with the first direction Z, and the pressure relief slot **102** is in communication with the pressure relief gap **101**. Based on this, the gas at the side portion of the insulating member **13** in the direction intersecting with the first direction Z can sequentially pass through the pressure relief gap **101** and the pressure relief slot **102** corresponding to the side portion and can thus flow to the pressure relief channel **103** and then to the pressure relief mechanism **123**.

[0108] In some possible designs, the pressure relief gap **101** is provided between one side of the insulating member **13** in the direction intersecting with the first direction Z and the housing **12**, and the pressure relief slot **102** is provided in the side of the insulating member **13** provided with the pressure relief gap **101** in the direction intersecting with the first direction Z. The pressure relief slot **102** is in communication with the pressure relief gap **101** on the corresponding side. In some possible designs, a pressure relief gap **101** is provided between each of two opposite sides of the insulating member **13** in the direction intersecting with the first direction Z and the housing **12**, and the insulating member **13** is provided with a pressure relief slot **102** at each of two opposite sides in the direction intersecting with the first direction Z. The pressure relief slots **102** at the two opposite sides of the insulating member **13** are in communication with the pressure relief gaps **101** at two opposite sides of the insulating member **13** in one-to-one correspondence.

[0109] The above-mentioned “direction intersecting with the first direction Z” may be the second direction Y or the third direction X indicated in the figures. That is, the second direction Y intersects with the first direction Z, and the third direction X intersects with the first direction Z. It should be explained that the second direction Y intersecting with the first direction Z means that the first direction Z and the second direction Y have an included angle greater than 0° and less than 180° formed therebetween, that is, the first direction Z is not parallel to the second direction Y. The first direction Z and the second direction Y may be perpendicular or not be perpendicular to each other. The first direction Z and the second direction Y may be directions that intersect with each other in the same plane, or may be directions in two non-coplanar planes, respectively, and the projection of the second direction Y on the plane in which the first direction Z is located may intersect with the first direction Z. The intersecting of the first direction Z and the third direction X may also be interpreted in the same way and will not be repeated again herein. The first direction Z, the second direction Y, and the third direction X are not in the same plane. As an example, as shown in FIGS. 3-8, the first direction Z is perpendicular to the second direction Y, the first direction Z is perpendicular to the third direction X, and the second direction Y is perpendicular to the third direction X.

[0110] In some possible designs, as shown in FIGS. 7 and 8 and in combination with the other figures, a pressure relief gap **101** is provided between a side portion of the insulating member **13** in the second direction Y and an inner side wall of the housing **12** in the second direction Y, and the pressure relief gap **101** is a first gap **1011**. The insulating member **13** is provided with a pressure

relief slot **102** at the side portion in the second direction Y, such that the pressure relief slot **102** can communicate with the pressure relief gap **101**. The pressure relief slot **102** is a first slot **1021**. In some possible designs, as shown in FIGS. **10** and **11** and in combination with the other figures, a pressure relief gap **101** is provided between a side portion of the insulating member **13** in the third direction X and an inner side wall of the housing **12** in the third direction X, and the pressure relief gap **101** is a second gap **1012**. The insulating member **13** is provided with a pressure relief slot **102** at the side portion in the third direction X, and the pressure relief slot **102** is a third slot **1022**.

[0111] The first direction Z is shown as the direction Z in the figures. In some cases, the first direction Z is the height direction of the battery cell **10**. In the case that the battery cell **10** is substantially square, as shown in FIG. **3**, the second direction Y may be the width direction of the battery cell **10**, and the third direction X may be the thickness direction of the battery cell **10**. Certainly, the second direction Y may also be the thickness direction of the battery cell **10**, and the third direction X may also be the width direction of the battery cell **10**. In the case that the battery cell **10** is substantially cylindrical, the second direction Y and the third direction X may be two intersecting diametrical directions of the battery cell **10**.

[0112] As shown in FIGS. **8** and **11**, in the direction intersecting with the first direction Z, the side portion of the insulating member **13** is spaced apart from the inner side wall of the housing body **121** to form a pressure relief gap **101**. Alternatively, in the direction intersecting with the first direction Z, the side portion of the insulating member **13** is spaced apart from an inner side wall of the end cover **122** to form a pressure relief gap **101**. Alternatively, in the direction intersecting with the first direction Z, the side portion of the insulating member **13** is spaced apart from the inner side wall of the end cover **122** to form a pressure relief gap **101**, and the side portion of the insulating member **13** is spaced apart from the inner side wall of the housing body **121** to form a pressure relief gap.

[0113] In the battery cell **10** provided in the embodiments of the present application, in the direction intersecting with the first direction Z, the pressure relief gap **101** is provided between the side portion of the insulating member **13** and the housing **12**, the insulating member **13** is provided with the pressure relief slot **102** at the side portion, the housing **12** is provided therein with the pressure relief channel **103** opposite to the pressure relief mechanism **123**, and the pressure relief slot **102** is in communication with the pressure relief gap **101** and the pressure relief channel **103**. In the case of the thermal runaway of the electrode assembly **11**, the high-temperature and high-pressure gas will be generated at the side portion of the electrode assembly **11** in the direction intersecting with the first direction Z, or the high-temperature and high-pressure gas generated by the electrode assembly **11** may flow to the side portion of the electrode assembly **11** in the direction intersecting with the first direction Z. As such, the high-temperature and high-pressure gas accumulates at the side portion of the electrode assembly **11** in the direction intersecting with the first direction Z, and the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** can sequentially pass through the pressure relief gap **101** and the pressure relief slot **102** to flow to the pressure relief channel **103** and thus to the pressure relief mechanism **123**, such that the gas can be released through the pressure relief mechanism **123**. In such a configuration, providing the pressure relief gap **101**, the pressure relief slot **102**, and the pressure relief channel **103** enables the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** to be released to the pressure relief mechanism **123** timely, so as to achieve timely pressure relief. As such, the problem of difficulties in timely releasing the high-temperature and high-pressure gas from the electrode assembly **11** caused by the insulating member **13** blocking the high-temperature and high-pressure gas can be ameliorated so as to ameliorate the problem of damage or even explosion of the electrode assembly **11** caused by the difficulties in timely pressure relief of the electrode assembly **11**, thereby imparting the battery cell **10** the advantage of timely pressure relief, which can suppress the thermal runaway and explosion of the battery cell **10** to a certain extent.

[0114] In some embodiments, referring to FIG. 4, an insulating film **14** is wrapped outside the electrode assembly **11**. The insulating film **14** is connected to the insulating member **13** to allow the insulating film **14** to remain in a standing state, thereby maintaining the insulation effect between the electrode assembly **11** and the housing **12**.

[0115] In some embodiments, referring to FIGS. 5-8 together, the pressure relief channel **103** is provided penetrating through the insulating member **13** in the first direction Z.

[0116] In such a configuration, one of the ends of the pressure relief channel **103** in the first direction Z is opposite to the electrode assembly **11**, and the other end of the pressure relief channel **103** in the first direction Z is opposite to the pressure relief mechanism **123**. In this way, in the case of the thermal runaway of the battery cell **10**, the high-temperature and high-pressure gas generated by the electrode assembly **11** can pass through the pressure relief channel **103** in the first direction Z and flow to the pressure relief mechanism **123** so as to be released to the outside of the battery cell **10** through the pressure relief mechanism **123**. Moreover, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** can sequentially pass through the pressure relief gap **101** and the pressure relief slot **102** to flow to the pressure relief channel **103** and then flow to the pressure relief mechanism **123** in the first direction Z. As such, the release of the high-temperature and high-pressure gas can be effectively achieved to achieve timely pressure relief of the battery cell **10**.

[0117] In some embodiments, referring to FIGS. 5-8 together and in combination with the other figures, the insulating member **13** includes a body part **131** and a first protrusion **132**. The first protrusion **132** is connected to the body part **131** and the first protrusion **132** extends beyond the body part **131** in the first direction Z toward the electrode assembly **11**. The pressure relief gap **101** includes a first gap **1011**. The above first gap **1011** is provided between the first protrusion **132** and the housing **12**. The pressure relief slot **102** includes a first slot **1021**. The first protrusion **132** is provided with the above first slot **1021**, and the first slot **1021** is in communication with the first gap **1011**. The pressure relief channel **103** includes a first channel **1031**. The first channel **1031** is provided in the first protrusion **132**, and the first channel **1031** is in communication with the first slot **1021** and disposed opposite to the pressure relief mechanism **123**.

[0118] The body part **131** and the first protrusion **132** each refer to a member made of an insulating material.

[0119] It may be appreciated that the first gap **1011** is provided between a side portion of the first protrusion **132** and the inner side wall of the housing **12** in the direction intersecting with the first direction Z. In the direction intersecting with the first direction Z, the first protrusion **132** is provided, at the side portion, with the first slot **1021** in communication with the first gap **1011**. "The direction intersecting with the first direction Z" may be the second direction Y in the figures or the third direction X in the figures.

[0120] It may be appreciated that the first protrusion **132** is opposite to the pressure relief mechanism **123** in the first direction Z. The first channel **1031** of the pressure relief channel **103** is formed in the first protrusion **132**, such that the first channel **1031** is opposite to the pressure relief mechanism **123** in the first direction Z.

[0121] In the case of the pressure relief channel **103** penetrating through the insulating member **13** in the first direction Z, the first channel **1031** may penetrate through the first protrusion **132** in the first direction Z, such that the first channel **1031** is opposite to the electrode assembly **11** in the first direction Z and opposite to the pressure relief mechanism **123** in the first direction Z.

[0122] In some possible designs, the first protrusion **132** extends beyond the body part **131** in the first direction Z toward the electrode assembly **11**, such that the first protrusion **132** may press against the electrode assembly **11** in the first direction Z. In this way, the electrode assembly **11** can be stably fixed in the housing **12** under the action of the pressure from the first protrusion **132**. In some possible designs, the first protrusion **132** may also not press against the electrode assembly **11**, and instead is spaced apart from the electrode assembly **11**. In this way, a gap between the

electrode assembly **11** and the first protrusion **132** provides a buffer for the high-temperature and high-pressure gas from the electrode assembly **11**, which in turn facilitates the release of the high-temperature and high-pressure gas through the electrode assembly **11** and the first protrusion **132** to the pressure relief mechanism **123**, ameliorating the problem of prolonged accumulation of the high-temperature and high-pressure gas in the electrode assembly **11**.

[0123] With the technical solution described above, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the direction intersecting with the first direction Z can enter the first slot **1021** of the first protrusion **132** through the first gap **1011** between the first protrusion **132** and the housing **12** and is then released to the pressure relief mechanism **123** through the first channel **1031** of the first protrusion **132**. As such, the release path of the high-temperature and high-pressure gas can be shortened to quickly realize the pressure relief of the high-temperature and high-pressure gas, thereby ameliorating the problem of difficulties in timely releasing the high-temperature and high-pressure gas caused by the first protrusion **132** blocking the high-temperature and high-pressure gas.

[0124] In some embodiments, referring to FIGS. 5-8 together and in combination with the other figures, the first protrusion **132** and the body part **131** are sequentially disposed in the second direction Y. In the third direction X, the first gap **1011** is provided between the side portion of the first protrusion **132** and the housing **12**, and the first protrusion **132** is provided with the first slot **1021** at the side portion. The first direction Z intersects with the second direction Y, the first direction Z intersects with the third direction X, and the second direction Y intersects with the third direction X.

[0125] The second direction Y is shown as the direction Y in the figures, and the third direction X is shown as the direction X in the figures.

[0126] The first protrusion **132** and the body part **131** being sequentially disposed in the second direction Y may be that the first protrusion **132** is provided in the middle of the body part **131** in the second direction Y, as shown in FIGS. 5 and 6, or it may also be that the first protrusion **132** is provided on the side portion of the body part **131** in the second direction Y, where specifically, the body part **131** is provided with a first protrusion **132** at one side or each of two opposite sides in the second direction Y.

[0127] A first gap **1011** is provided between one side or each of two opposite sides of the first protrusion **132** in the third direction X and the inner side wall of the housing **12**, and the first protrusion **132** is provided with a first slot **1021** at one side or each of two opposite sides in the third direction X. Also, the first slot **1021** is in communication with the first gap **1011** and the first channel **1031** in the third direction X.

[0128] With the technical solution described above, the first gap **1011** and the first slot **1021** are provided at the side portion of the first protrusion in the third direction X, such that the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X can sequentially pass through the first gap **1011** and the first slot **1021** to flow to the first channel **1031** and thus to the pressure relief mechanism **123**. As such, the release of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X can be effectively achieved to achieve timely pressure relief of the battery cell **10**.

[0129] In some embodiments, referring to FIGS. 5 and 6 together and in combination with the other figures, the first protrusion **132** includes two first walls **1321**. The two first walls **1321** are spaced apart from each other on the body part **131** in the second direction Y to form the first channel **1031** in a spaced manner. The first slot **1021** is formed in side portions of the two first walls **1321** in the third direction X.

[0130] It may be appreciated that the two first walls **1321** are two opposite side walls of the first channel **1031** in the second direction Y. When the first protrusion **132** is disposed in the middle of the body part **131** in the second direction Y, the two first walls **1321** may be disposed directly on the body part **131**. When the first protrusion **132** is disposed at an end of the body part **131** in the

second direction Y, one of the first walls **1321** may be disposed directly on the body part **131** and the other first wall **1321** may be indirectly connected to the body part **131** by another component, such that the two first walls **1321** can be spaced apart in the second direction Y to form the first channel **1031**.

[0131] When one side of the first protrusion **132** in the third direction X is spaced apart from the housing **12** to form the first gap **1011**, one side of each of the two first walls **1321** in the third direction X is spaced apart from the housing **12** to form the first gap **1011**. When each of the two opposite sides of the first protrusion **132** in the third direction X forms the first gap **1011** together with the housing **12**, the two opposite sides of each of the two first walls **1321** in the third direction X are spaced apart from the housing **12** to form the first gaps **1011**.

[0132] When the first protrusion **132** is provided with a first slot **1021** at one side in the third direction X, the first slot **1021** above is formed at one side of the two first walls **1321** in the third direction X. When the first protrusion **132** is provided with a first slot **1021** at each of two opposite sides in the third direction X, a first slot **1021** is formed at each of two opposite sides of the two first walls **1321** in the third direction X.

[0133] The first walls **1321** may abut against the electrode assembly **11** in the first direction Z, and the first walls **1321** may also be spaced apart from the electrode assembly **11** in the first direction Z.

[0134] By forming the first channel **1031** by the two first walls **1321**'s being spaced apart from each other in the second direction Y, the first channel **1031** may be provided penetrating through the insulating member **13** in the first direction Z, and the first channel **1031** has a larger dimension in a cross section perpendicular to the first direction Z, which facilitates the high-temperature and high-pressure gas's passing through the first channel **1031** to flow to the pressure relief mechanism **123** and thereby to achieve timely pressure relief of the battery cell **10**. Also, the ends of the two first walls **1321** in the third direction X may be spaced apart to form the first slot **1021**, such that the first slot **1021** has a large dimension, which also facilitates the passing of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X through the first gap **1011** and the first slot **1021** to enter the first channel **1031**, thus facilitating the timely pressure relief of the electrode assembly **11**.

[0135] In some embodiments, referring to FIGS. 5 and 6 together and in combination with the other figures, the first protrusion **132** further includes a second wall **1322**. The second wall **1322** is disposed between the two first walls **1321** and at the side portions of the first walls **1321** in the third direction X. The first walls **1321** and the second wall **1322** define the first channel **1031**, and the first slot **1021** is formed in the second wall **1322**.

[0136] The second wall **1322** refers to a side wall of the first channel **1031** in the third direction X

[0137] A second wall **1322** is disposed at one side of the first wall **1321** in the third direction X. Alternatively, a second wall **1322** is disposed at each of two opposite sides of the first wall **1321** in the third direction X.

[0138] The second wall **1322** may abut against the electrode assembly **11** in the first direction Z, or the second wall **1322** may also be spaced apart from the electrode assembly **11** in the first direction Z.

[0139] The first slot **1021** is provided penetrating through the second wall **1322** in the third direction X, such that the first slot **1021** is in communication with both the first gap **1011** and the first channel **1031** in the third direction X. Also, in some possible designs, the first slot **1021** is provided with an opening on a side facing the electrode assembly **11** in the first direction Z. In some possible designs, the first slot **1021** is provided with an opening on a side facing away from the electrode assembly **11** in the first direction Z. In some possible designs, no opening is provided on the two opposite sides of the first slot **1021** in the first direction Z.

[0140] With the technical solution described above, the second wall **1322** is provided between the two first walls **1321**, and the first slot **1021** is formed in the second wall **1322**. As such, the

insulating film **14** may extend to the second wall **1322** to be fixed to the second wall **1322**, such that the insulating film **14** remains in the standing state, thereby maintaining the insulation effect between the electrode assembly **11** and the housing **12**.

[0141] In some embodiments, referring to FIGS. 5-6 together and in combination with the other figures, the first protrusion **132** further includes a third wall **1323**. The third wall **1323** is disposed between the two first walls **1321** and opposite to the pressure relief mechanism **123** in the first direction Z. The first channel **1031** includes a first space **10311** and second slots **10312**. The first walls **1321** and the side of the third wall **1323** distal to the electrode assembly **11** in the first direction Z define the first space **10311**, and the first space **10311** is in communication with the first slot **1021** and opposite to the pressure relief mechanism **123** in the first direction Z. The second slots **10312** penetrate through the third wall **1323** in the first direction Z and are in communication with the first space **10311**.

[0142] It may be appreciated that the first walls **1321** are the side walls of the first space **10311** in the second direction Y, the second wall **1322** is the side wall of the first space **10311** in the third direction X, and the third wall **1323** is the side wall of the first space **10311** facing the electrode assembly **11** in the first direction Z, and the first space **10311** is also disposed opposite to the pressure relief mechanism **123** in the first direction Z.

[0143] With the technical solution described above, the first protrusion **132** defines the first space **10311** and the second slots **10312**, the high-temperature and high-pressure gas generated by the electrode assembly **11** can pass through the second slots **10312** in the first direction Z to enter the first space **10311**, and the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X can sequentially pass through the first gap **1011** and the first slot **1021** to enter the first space **10311**. As such, the high-temperature and high-pressure gas can be buffered in the first space **10311** and then flow to the pressure relief mechanism **123**, allowing the high-temperature and high-pressure gas in the battery cell **10** to rush out through the pressure relief mechanism **123** to realize pressure relief.

[0144] In some embodiments, referring to FIGS. 5-8 together and in combination with the other figures, the third wall **1323** is disposed at ends of the first walls **1321** facing the electrode assembly **11** in the first direction Z and is configured to abut against the electrode assembly **11** in the first direction Z.

[0145] With the technical solution described above, the third wall **1323** may abut against the electrode assembly **11** in the first direction Z. As such, the area of the first protrusion **132** for contact with the electrode assembly **11** can be increased, such that the first protrusion **132** can abut against the electrode assembly **11** very stably, thus fixing the electrode assembly **11** in the housing **12** and allowing for reduced damages to the electrode assembly **11** caused by the first protrusion **132**.

[0146] In some embodiments, referring to FIGS. 5 and 6 together and in combination with the other figures, the body part **131** includes a plurality of sub-parts **1311**, and the plurality of sub-parts **1311** are sequentially disposed in the second direction Y. Also, the first protrusion **132** is disposed between two adjacent sub-parts **1311**.

[0147] It may be appreciated that the first protrusion **132** and the plurality of sub-parts **1311** are sequentially disposed in the second direction Y, realizing the effect of disposing the first protrusion **132** and the body part **131** in the second direction Y.

[0148] In such a configuration, the first protrusion **132** is disposed in the middle of the body part **131** in the second direction Y, and the first protrusion **132** is thus directly opposite to the middle of the electrode assembly **11** in the second direction Y. In some cases, when the electrode assembly **11** experiences thermal runaway, the high-temperature and high-pressure gas generated by the electrode assembly **11** is concentrated in large amount in the middle of the electrode assembly **11**. As such, the high-temperature and high-pressure gas generated by the electrode assembly **11** can quickly flow to the first channel **1031** to be released through the pressure relief mechanism **123**.

This helps to achieve the effect of timely pressure relief of the battery cell **10**.

[0149] In some embodiments, the first slot **1021** and the first protrusion **132** satisfy the following relationship:

$S1 \geq S2 \times 1/6$, [0150] where on one of the side surfaces of the first protrusion **132** in the third direction X, S1 is the area of the first slot **1021**, and S2 is the area of the first protrusion **132**.

[0151] On one of the side surfaces of the first protrusion **132** in the third direction X, one or more first slots **1021** may be provided. When the number of first slots **1021** is one, S1 is the area of the first slot **1021**. When the number of first slots **1021** is plural, S1 is the sum of the areas of the plurality of first slots **1021**, i.e., the sum of the areas of all of the first slots **1021** on the side surface.

[0152] By setting $S1 \geq S2 \times 1/6$, the area of the first slot **1021** is not too small, such that the first slot **1021** can allow the high-temperature and high-pressure gas to flow through timely, which facilitates the high-temperature and high-pressure gas' entering the first channel **1031** through the first slot **1021** for pressure relief.

[0153] It should be noted that the gas pressures and temperatures at the side portion of the electrode assembly **11** in the third direction X and at the first channel **1031** may be measured when the thermal runaway of the battery cell **10** is simulated. When the pressure and temperature at the side portion of the electrode assembly **11** are substantially the same as the gas pressure and temperature at the first channel **1031**, it can be considered that the first channel **1031** is in communication with the side portion of the electrode assembly **11**. By setting $S1 \geq S2 \times 1/6$, the speed of communication between the first channel **1031** and the side portion of the electrode assembly **11** can be increased, and the pressure relief of the side portion of the electrode assembly **11** can be achieved quickly.

[0154] In some embodiments, $S1 \geq S2 \times 1/2$.

[0155] By setting $S1 \geq S2 \times 1/2$, the first slot **1021** can have a large area proportion on the first protrusion **132**, such that the effect of rapid release of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** to the pressure relief mechanism **123** can be achieved.

[0156] In some embodiments, the first protrusion **132** is provided on a side portion of the body part **131** in the second direction Y. In the second direction Y, the first gap **1011** is provided between a side portion of the first protrusion **132** distal to the body part **131** and the housing **12**, and the first protrusion **132** is provided with the first slot **1021** at the side portion, where the first direction Z intersects with the second direction Y.

[0157] The body part **131** is provided with a first protrusion **132** at one side in the second direction Y, or the body part **131** is provided with a first protrusion **132** at each of two opposite sides in the second direction Y.

[0158] The first gap **1011** is provided between a side of the first protrusion **132** distal to the body part **131** in the second direction Y and the housing **12**, and the first protrusion **132** is provided with the first slot **1021** at the side distal to the body part **131** in the second direction Y, and the first slot **1021** is in communication with the first gap **1011** in the second direction Y.

[0159] In such a configuration, the high-temperature and high-pressure gas on the side portion of the electrode assembly **11** in the second direction Y can sequentially pass through the first gap **1011** and the first slot **1021** to enter the first channel **1031** and thereby to flow to the pressure relief mechanism **123** through the first channel **1031**. This enables the effect of timely release of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the second direction Y to achieve timely pressure relief of the electrode assembly **11**.

[0160] In some embodiments, referring to FIGS. 9-12 together and in combination with the other figures, FIG. 9 shows a schematic perspective view of the insulating member **13** of the battery cell **10** according to some other embodiments of the present application, FIG. 10 is a cross-sectional view of the battery cell **10** in which the insulating member **13** is located in FIG. 9, FIG. 11 is a

partial enlarged view of FIG. 10, and FIG. 12 is a partial enlarged view of FIG. 10. The insulating member 13 includes the body part 131. In the first direction Z, the body part 131 is spaced apart from the electrode assembly 11 to form a second space 104, and the second space 104 is in communication with the pressure relief channel 103.

[0161] The pressure relief channel 103 may be formed in the body part 131. Alternatively, the pressure relief channel 103 is formed in the first protrusion 132. Alternatively, a pressure relief channel 103 is formed in each of the body part 131 and the first protrusion 132.

[0162] In some embodiments, referring to FIGS. 9-12 together and in combination with the other figures, the insulating member 13 further includes a second protrusion 133. In the first direction Z, the second protrusion 133 is disposed on the side of the body part 131 facing the electrode assembly 11. The pressure relief gap 101 includes a second gap 1012, and the pressure relief slot 102 includes a third slot 1022. In the direction intersecting with the first direction Z, the second protrusion 133 is spaced apart from the housing 12 to form the second gap 1012. In the direction intersecting with the first direction Z, the second protrusion 133 is provided with the third slot 1022. The third slot 1022 is in communication with the second gap 1012 and the second space 104.

[0163] The above-mentioned “direction intersecting with the first direction Z” may be the second direction Y or the third direction X. Specifically, in some possible designs, the above third slot 1022 is formed in the side portion of the second protrusion 133 in the second direction Y, the side portion of the second protrusion 133 in the second direction Y is spaced apart from the housing 12 to form the above second gap 1012, the third slot 1022 is in communication with the second gap 1012 in the second direction Y, and the third slot 1022 is in communication with the second space 104. Based on this, the high-temperature and high-pressure gas at the side portion of the electrode assembly 11 in the second direction Y can sequentially pass through the corresponding second gap 1012 and third slot 1022 to flow to the second space 104. In some possible designs, the above third slot 1022 is formed in the side portion of the second protrusion 133 in the third direction X, and the side portion of the second protrusion 133 in the third direction X is spaced apart from the housing 12 to form the above second gap 1012. The above second gap 1012 is in communication with the above third slot 1022 in the third direction X, and the third slot 1022 is in communication with the second space 104. Based on this, the high-temperature and high-pressure gas on the side portion of the electrode assembly 11 in the third direction X can sequentially pass through the corresponding second gap 1012 and third slot 1022 to flow to the second space 104.

[0164] The second protrusion 133 may abut against the electrode assembly 11 in the first direction Z, such that the electrode assembly 11 can be fixed in the battery cell 10. The second protrusion 133 may also be spaced apart from the electrode assembly 11 in the first direction Z, such that a gap between the second protrusion 133 and the electrode assembly 11 allows the high-temperature and high-pressure gas to pass through to enter the second space 104 and thus flow to the pressure relief channel 103 for pressure relief through the pressure relief mechanism 123.

[0165] With the technical solution described above, in the case where there is high-temperature and high-pressure gas at the side portion of the electrode assembly 11 in the direction intersecting with the first direction Z, the high-temperature and high-pressure gas can sequentially pass through the first gap 1011 and the second gap 1012 to flow to the pressure relief channel 103, or can pass through the second gap 1012 and the third slot 1022 to flow to the second space 104 and then to the pressure relief channel 103, thus being released to the pressure relief mechanism 123. This helps to achieve the effect of timely pressure relief of the electrode assembly 11.

[0166] In some embodiments, referring to FIGS. 9-13 together and in combination with the other figures, FIG. 13 is a partial enlarged view of FIG. 9. The insulating member 13 includes the body part 131 and the first protrusion 132, and the first protrusion 132 and the body part 131 are sequentially disposed in the second direction Y. In the first direction Z, the first protrusion 132 extends beyond the body part 131 in the direction towards the electrode assembly 11. In the first direction Z, the body part 131 is spaced apart from the electrode assembly 11 to form the second

space **104**, and the second space **104** is in communication with the pressure relief channel **103**. The pressure relief channel **103** includes the first channel **1031**. The first channel **1031** is provided in the first protrusion **132** and disposed opposite to the pressure relief mechanism **123**. The first protrusion **132** is provided with fourth slots **105** at the side portion in the second direction Y, and the fourth slots **105** are in communication with the second space **104** and the first channel **1031**. The first direction Z intersects with the second direction Y.

[0167] The second space **104** referred to in this embodiment is the same as the second space **104** referred to in the above embodiments, and both of the spaces are formed by the body part **131**'s being spaced part from the electrode assembly **11** in the first direction Z.

[0168] The first protrusion **132** and the body part **131** are sequentially disposed in the second direction Y, and the second protrusion **133** is disposed on the side of the body part **131** facing the electrode assembly **11** in the first direction Z. As such, the first protrusion **132** is spaced apart from the second protrusion **133** in the second direction Y, and the first protrusion **132** and the second protrusion **133** have the second space **104** therebetween.

[0169] With the technical solution described above, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** can flow to the third slot **1022** of the second protrusion **133** through the second gap **1012** between the second protrusion **133** and the housing **12** to flow to the second space **104** and thus to the first channel **1031** through the fourth slots **105** in communication with the second space **104**, such that the gas can flow to the pressure relief mechanism **123** to realize the timely release of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11**. Also, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** can directly enter the second space **104** and thus flow to the first channel **1031** through the fourth slots **105** in communication with the second space **104**, such that the gas can flow to the pressure relief mechanism **123** to realize the timely release of the high-temperature and high-pressure gas at the side portion of the electrode assembly **11**. In such a configuration, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the direction intersecting with the first direction Z can flow to the second space **104** through a plurality of pathways, and thus flow to the first channel **1031** through the fourth slots **105** to be released through the pressure relief mechanism **123**, such that the battery cell **10** has a plurality of channels that can release the high-temperature and high-pressure gas, which contributes to the timely pressure relief of the battery cell **10**.

[0170] In some embodiments, referring to FIGS. 9-13 together and in combination with the other figures, the insulating member **13** includes the body part **131**, and the body part **131** is spaced apart from the electrode assembly **11** in the first direction Z to form the second space **104**. The pressure relief channel **103** includes the second channel **1032**, and the second channel **1032** includes a fifth groove **10321** and sixth slots **10322**. The fifth groove **10321** is provided on the side of the body part **131** distal to the electrode assembly **11** in the first direction Z and is opposite to the pressure relief mechanism **123**. The sixth slots **10322** penetrate through the bottom of the fifth groove **10321** in the first direction Z and are in communication with the second space **104**.

[0171] In such a configuration, the high-temperature and high-pressure gas in the second space **104** may not only flow to the first channel **1031** through the fourth slots **105**, but also flow to the sixth slots **10322** through the fifth groove **10321**, and is then released to the pressure relief mechanism **123**. In this way, the high-temperature and high-pressure gas in the second space **104** can be released to the pressure relief mechanism **123** through a plurality of pathways, thus facilitating the timely pressure relief of the battery cell **10**.

[0172] In some embodiments, the fifth groove **10321** of the second channel **1032** is in communication with the first space **10311** of the first channel **1031**, thus facilitating the buffer and release of the high-temperature and high-pressure gas.

[0173] Based on the above concept, referring to FIG. 2, an embodiment of the present application further provides a battery **100**. The battery **100** includes a battery cell **10**. The battery cell **10** in this

embodiment is the same as the battery cell **10** in the previous embodiment. Reference may be made to the relevant descriptions of the battery cell **10** in the previous embodiment for details, which are not described again herein.

[0174] According to the battery **100** provided in the embodiment of the present application, the use of the battery cell **10** referred to above can ameliorate the problem of difficulties in timely releasing the high-temperature and high-pressure gas from the electrode assembly **11** caused by the insulating member **13** blocking the high-temperature and high-pressure gas, so as to ameliorate the problem of damage or even explosion of the electrode assembly **11** caused by the difficulties in timely pressure relief of the electrode assembly **11**, thereby imparting the battery **100** the advantage of timely pressure relief.

[0175] Based on the above concept, referring to FIG. **1**, an embodiment of the present application further provides an electric device. The electric device includes the battery cell **10** or the battery **100**. The battery cell **10** and the battery **100** in this embodiment are the same as the battery cell **10** and the battery **100** in the previous embodiment. Reference may be made to the relevant descriptions of the battery cell **10** and the battery **100** in the previous embodiment for details, which are not described again herein.

[0176] According to the electric device provided in the embodiment of the present application, the use of the battery cell **10** or battery **100** referred to above can ameliorate the problem of difficulties in timely releasing the high-temperature and high-pressure gas from the electrode assembly **11** caused by the insulating member **13** blocking the high-temperature and high-pressure gas, so as to ameliorate the problem of damage or even explosion of the electrode assembly **11** caused by the difficulties in timely pressure relief of the electrode assembly **11**, thereby imparting the battery **100** the advantage of timely pressure relief.

[0177] As an embodiment of the present application, as shown in FIGS. **5-8**, the battery cell **10** includes a housing **12**, an electrode assembly **11**, and an insulating member **13**. The housing **12** is provided with a pressure relief mechanism **123** at an end in the first direction Z, and the insulating member **13** is disposed at an end of the electrode assembly **11** proximal to the pressure relief mechanism **123**. The insulating member **13** includes a body part **131** and a first protrusion **132**. The first protrusion **132** is disposed in the middle of the body part **131** in the second direction Y and is sequentially disposed with the body part **131** in the second direction Y. In the third direction X, a side portion of the first protrusion **132** is spaced apart from the housing **12** to form a first gap **1011**, and the first protrusion **132** is provided, at the side portion, a first slot **1021** in communication with the first gap **1011**. A first channel **1031** in communication with the first slot **1021** penetrates through the first protrusion **132** in the first direction Z. Based on this, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X can sequentially pass through the first gap **1011** and the first slot **1021** to enter the first channel **1031** and thereby to be released to the pressure relief mechanism **123**.

[0178] As shown in FIGS. **9-13**, in the first direction Z, the body part **131** is spaced apart from the electrode assembly **11** to form a second space **104**. The first protrusion **132** is provided with fourth slots **105** at the side portion in the second direction Y, and the fourth slots **105** are in communication with the first channel **1031**. A second channel **1032** is formed in the body part **131**, and the second channel **1032** includes a fifth groove **10321** and sixth slots **10322**. In the first direction Z, the body part **131** is provided with the fifth groove **10321** at the side distal to the electrode assembly **11**, and the sixth slots **10322** are formed in the bottom of the fifth groove **10321** by penetrating through it in the first direction Z and the sixth slots **10322** are in communication with the second space **104**. Based on this, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the third direction X can flow to the second space **104** through the gap between the electrode assembly **11** and the housing **12**, and can thus sequentially pass through the fourth slots **105** and the first channel **1031** to be released to the pressure relief mechanism **123**, or can sequentially pass through the fifth groove **10321** and the sixth slots **10322**

to be released to the pressure relief mechanism **123**.

[0179] As shown in FIGS. **9-13**, the insulating member **13** further includes a second protrusion **133** disposed on the side of the body part **131** facing the electrode assembly **11** in the first direction Z. The second protrusion **133** is spaced apart from the first protrusion **132** in the second direction Y, and the first protrusion **132** and the second protrusion **133** have the above second space **104** therebetween. A third slot **1022** is provided penetrating through the second protrusion **133** in the second direction Y, and the side portion of the second protrusion **133** in the second direction Y is spaced apart from the housing **12** to form a second gap **1012**. The second gap **1012** is in communication with the third slot **1022**, and the third slot **1022** is in communication with the second space **104**. Based on this, the high-temperature and high-pressure gas at the side portion of the electrode assembly **11** in the second direction Y can sequentially pass through the second gap **1012** and third slot **1022** to flow to the second space **104**, and can thus sequentially pass through the fourth slots **105** and the first channel **1031** to be released to the pressure relief mechanism **123**, or can sequentially pass through the fifth groove **10321** and the sixth slots **10322** to be released to the pressure relief mechanism **123**.

[0180] The above are only optional embodiments of the present application and are not intended to limit the present application. For those skilled in the art, the present application can be modified and varied. Any modification, equivalent substitution, improvement, and the like made within the spirit and principle of the present application shall all fall within the scope of the claims of the present application.

Claims

1. A battery cell, comprising: a housing provided with a pressure relief mechanism at an end in a first direction (Z); an electrode assembly disposed in the housing; and an insulating member disposed in the housing and located at an end of the electrode assembly proximal to the pressure relief mechanism, wherein in a direction intersecting with the first direction (Z), a pressure relief gap is provided between a side portion of the insulating member and the housing, the insulating member is provided, at the side portion, with a pressure relief slot in communication with the pressure relief gap, and the housing is provided therein with a pressure relief channel in communication with the pressure relief slot and disposed opposite to the pressure relief mechanism.
2. The battery cell according to claim 1, wherein the pressure relief channel is provided penetrating through the insulating member in the first direction (Z).
3. The battery cell according to claim 1, wherein the insulating member comprises a body part and a first protrusion connected to the body part, wherein the first protrusion extends beyond the body part in the first direction (Z); the pressure relief gap comprises a first gap provided between the first protrusion and the housing, the pressure relief slot comprises a first slot provided in the first protrusion and in communication with the first gap, the pressure relief channel comprises a first channel provided in the first protrusion, and the first channel is in communication with the first slot and is opposite to the pressure relief mechanism.
4. The battery cell according to claim 3, wherein the first protrusion and the body part are sequentially disposed in a second direction (Y); in a third direction (X), the first gap is provided between a side portion of the first protrusion and the housing, and the first protrusion is provided with the first slot at the side portion; the first direction (Z), the second direction (Y), and the third direction (X) intersect with each other, and the first direction (Z), the second direction (Y), and the third direction (X) are not in a same plane.
5. The battery cell according to claim 4, wherein the first protrusion comprises: two first walls spaced apart from each other on the body part in the second direction (Y) to form the first channel, the first slot being formed in side portions of the two first walls in the third direction (X).
6. The battery cell according to claim 5, wherein the first protrusion further comprises: a second

wall disposed between the two first walls and at the side portions of the first walls in the third direction (X), the first walls and the second wall defining the first channel, and the first slot being formed in the second wall.

7. The battery cell according to claim 6, wherein the first protrusion further comprises a third wall, the third wall being disposed between the two first walls and opposite to the pressure relief mechanism; the first channel comprises: a first space in communication with the first slot and opposite to the pressure relief mechanism, the first walls and a side of the third wall distal to the electrode assembly in the first direction (Z) defining the first space; and second slots penetrating through the third wall in the first direction (Z) and in communication with the first space.

8. The battery cell according to claim 7, wherein the third wall is disposed at ends of the first walls facing the electrode assembly in the first direction (Z) and is configured to abut against the electrode assembly in the first direction (Z).

9. The battery cell according to claim 4, wherein the body part comprises a plurality of sub-parts sequentially disposed in the second direction (Y), and the first protrusion is disposed between two adjacent sub-parts.

10. The battery cell according to claim 4, wherein the first slot and the first protrusion satisfy the following relationship:

$S1 \geq S2 \times 1/6$, wherein on one of side surfaces of the first protrusion in the third direction (X), S1 is an area of the first slot, and S2 is an area of the first protrusion.

11. The battery cell according to claim 10, wherein $S1 \geq S2 \times 1/2$.

12. The battery cell according to claim 3, wherein the first protrusion is disposed on a side portion of the body part in the second direction (Y); in the second direction (Y), the first gap is provided between a side portion of the first protrusion distal to the body part and the housing, and the first protrusion is provided with the first slot at the side portion; the first direction (Z) intersects with the second direction (Y).

13. The battery cell according to claim 1, wherein the insulating member comprises the body part and a second protrusion; in the first direction (Z), the second protrusion is disposed on a side of the body part facing the electrode assembly, and the body part is spaced apart from the electrode assembly to form a second space in communication with the pressure relief channel; the pressure relief gap comprises a second gap, and the pressure relief slot comprises a third slot in communication with the second gap; in a direction intersecting with the first direction (Z), the second protrusion is spaced apart from the housing to form the second gap, and the second protrusion is provided with the third slot in communication with the second space.

14. The battery cell according to claim 1, wherein the insulating member comprises the body part and the first protrusion, and the first protrusion and the body part are sequentially disposed in the second direction (Y); in the first direction (Z), the first protrusion extends beyond the body part toward the electrode assembly, and the body part is spaced apart from the electrode assembly to form the second space in communication with the pressure relief channel; the pressure relief channel comprises the first channel provided in the first protrusion and opposite to the pressure relief mechanism, and the first protrusion is provided, at a side portion in the second direction (Y), with fourth slots (**105**) in communication with the second space and the first channel; the first direction (Z) intersects with the second direction (Y).

15. The battery cell according to claim 1, wherein the insulating member comprises the body part, and the body part is spaced apart from the electrode assembly in the first direction (Z) to form the second space; the pressure relief channel comprises a second channel, and the second channel comprises: a fifth groove provided on a side of the body part distal to the electrode assembly in the first direction (Z) and opposite to the pressure relief mechanism; and sixth slots penetrating through a bottom of the fifth groove in the first direction (Z) and in communication with the second space.

16. A battery, comprising the battery cell according to claim 1.

- 17.** An electric device, comprising the battery cell according to claim 1.
 - 18.** An electric device, comprising the battery according to claim 16.
-